



UNIVERSITY
OF THE PEOPLE



Power Plant Fitness: Design of an IoT-Integrated Member Retention and Energy Management System

A Capstone Project Report of MSIT 5910

Submitted by:

Ryan Bains-Jordan (S491861)

For the partial fulfillment of the requirements for the degree of

Master of Science in Information Technology

Supervised by:

Dr. Shabia Shabir

Department of CS & MSIT

University of the People, Pasadena, CA 91101, United States

AY2026-T4, June, 2026



ABSTRACT

The fitness industry faces a persistent member retention crisis, with research indicating that 40–65% of gym members disengage within their first six months due to insufficient motivational infrastructure. This project proposes and documents the design of Power Plant Fitness: a fictional IoT-integrated business management system that addresses this gap by converting the physical energy members generate during exercise into a redeemable digital currency called the Watt Wallet.

The project follows a documentation-driven development methodology, producing a system architecture diagram, high-fidelity UI mockups, and a functional Python implementation of the Wattage Credit Normalization Engine, which is the algorithm that calculates fair and equitable reward credits. The normalization engine rewards consistent gym attendance through a single behavioral modifier, ensuring that members who visit frequently earn meaningfully more than infrequent visitors, regardless of athletic output. The algorithm was validated through a 15-test PyTest suite, achieving a 100% pass rate, and deployed to AWS Lambda behind an API Gateway endpoint, where live integration and system testing confirmed end-to-end response times of 67–392ms depending on cold and warm execution states.

Key findings demonstrate that algorithmic fairness requires deliberate simplicity: focusing on a single, ungameable consistency factor outperforms more complex multi-factor approaches that introduce compounding penalties and gaming vulnerabilities. The project contributes a transferable reward model, a grace period mechanism for new members, and an ethical framework for evaluating behavioral metrics in IoT-integrated engagement systems.

Keywords: IoT, member retention, gamification, AWS Lambda, normalization algorithm, fitness technology



UNIVERSITY
OF THE PEOPLE



ACKNOWLEDGEMENTS

The author would like to express sincere gratitude to Dr. Shabia Shabir and all other instructors in the MSIT Program for their guidance and supervision throughout the degree program. Appreciation is also extended to the Program Advisors, who were available to answer questions about the degree program and to offer support when issues arose, as well as to the University of the People for providing an accessible, flexible, and rigorous academic environment that made the pursuit of a Master of Science in Information Technology possible.



LIST OF TABLES

Table 1. Resource Justification Table	16
Table 2. Energy-generating exercise machine I/O and requirements	17
Table 3. Energy-consuming exercise machine I/O and requirements	18
Table 4. RFID wearables and smartphone NFC I/O and requirements	18
Table 5. Machine-level touchscreen interface I/O and requirements	19
Table 6. Local battery and energy management I/O and requirements	19
Table 7. Administrative dashboard I/O and requirements	20
Table 8. Power Plant Fitness mobile app I/O and requirements	20
Table 9. Centralized cloud backend I/O and requirements	21



LIST OF FIGURES

Figure 1. Stakeholder Power/Interest Grid	9
Figure 2. System Architecture Diagram	15
Figure 3. Input/Output Demo for Samantha	23
Figure 4. Input/Output Demo for Jacob	24
Figure 5. Integration Test 1	26
Figure 6. Integration Test 2	27
Figure 7. Integration Test 3	27
Figure 8. Algorithm performance test	29
Figure 9. Tap-in screen	30
Figure 10. Home screen	31
Figure 11. Evaluation metrics and results	31
Figure 12. Monthly earnings comparison from demo	32



CONTENTS

ABSTRACT.....	2
ACKNOWLEDGEMENTS.....	3
LIST OF TABLES.....	4
LIST OF FIGURES	5
CONTENTS.....	6
INTRODUCTION	8
1.1 Background of the Study	8
Stakeholders.....	8
Problem Statement.....	9
1.3 Objectives of the Study	10
Project Goals (SMART Framework).....	10
Expected Outcomes	11
1.4 Scope and Limitations.....	11
1.5 Report Organization.....	12
LITERATURE REVIEW	13
2.1 Overview of Related Work	13
2.2 Comparative Analysis of Existing Systems	13
2.3 Research Gap Identification.....	14
SYSTEM DESIGN	15
3.1 System Architecture.....	15
3.2 Methodologies and Tools Used.....	16
3.3 Data Flow and System Components	17
Energy-Generating Exercise Machines.....	17
Energy-Consuming Exercise Machines	17
RFID Wearables and Smartphone NFC.....	18
Machine-Level Touchscreen Interface.....	19
Local Battery and Energy Management System.....	19
Administrative Dashboard	20
Power Plant Fitness Mobile App	20
Centralized Cloud Backend	20



3.4 Design Diagrams (DFD, UML, ER Diagram)	21
RESULTS	22
4.1 Implementation Overview.....	22
4.2 Output and Screenshots.....	22
4.3 Evaluation Metrics	24
Unit Tests	24
Integration Tests.....	26
System Tests	28
Algorithm Performance.....	29
Qualitative Metric (Interface Usability).....	29
Results Summary	31
DISCUSSION	33
5.1 Interpretation of Results.....	33
5.2 Comparison with Literature	33
5.3 Implications and Limitations	34
Selected Deployment Approach.....	34
Configuration Management	34
Limitations	35
5.4 Future Work.....	35
CONCLUSION.....	37
REFERENCES	39
APPENDIX.....	42
Figure A1	42
Figure A2	43
Figure A3	44



INTRODUCTION

1.1 Background of the Study

Many people believe in practicing physical fitness frequently, but with any activity that requires effort without an immediate reward, many lose interest quickly. For some, investing in a fitness club or gym membership is enough motivation to devote time and energy to exercise at their facilities, but for many, that is not enough. One study from the Journal of Sports Science & Medicine found that 40-65% of fitness memberships are canceled within the first six months, primarily due to a lack of sustained motivation and the absence of meaningful extrinsic incentives (Gjestvang et al., 2023). While long-term goals such as muscle growth or improved health are valuable, the results are often too gradual to prevent frustration and demotivation. Technology can be a valid solution to this problem, but most solutions focus heavily on passive monitoring, such as collecting real-time vitals during physical activity, guided workouts, nutrition, and activity tracking (Fabbrizio et al., 2023). The proposed solution to this problem falls within the IT domains of IoT systems architecture, data management, human-computer interaction (HCI), and business information systems.

Stakeholders

The stakeholders are vital to determining this solution because they can influence whether the project succeeds and the complexity of the result (Oguz, 2025). For that reason, the stakeholders who benefit most from this solution should be prioritized. In this case, gym members are the primary stakeholders because they drive the gym's financial success. This project focuses on membership retention by offering a compelling, measurable reason to return: an extrinsic reward system for sustained effort. Stakeholders within the organization include gym operators and staff, the second key stakeholder group. High dropout rates translate directly into lost revenue, and a system that improves retention has clear financial implications for the business. Finally, the environment is a third potential stakeholder given the project's sustainability dimension. To better understand the project's success factor in meeting stakeholder expectations, it is beneficial to recognize their power and interest in the project to determine the best strategy for engaging with them (Oguz, 2025). A Stakeholder Power/Interest Grid is demonstrated below.

Power	High	Gym staff, operation teams, and organization employees	Gym members and prospective members
	Low	The environment	Anyone interested in fitness with extrinsic rewards and an ecological impact
		Low	High
		Interest	

Figure 1. Stakeholder Power/Interest Grid

Problem Statement

The fitness industry suffers from a significant retention crisis. Those with gym and fitness club memberships often disengage with their routines within the first six months due to an insufficient motivational structure. Current IoT fitness technologies prioritize passive self-monitoring, milestones, and progress tracking but lack the extrinsic, reward-based incentives necessary to sustain long-term engagement (Fabbrizio et al., 2023). No existing platform directly connects a member's physical energy output to a personalized, redeemable reward economy, while simultaneously directing that energy to a positive ecological impact. This project proposes designing a unified IoT-integrated business management system for a fictional gym called "Power Plant Fitness" that addresses this gap by using smart exercise equipment to serve as both microgenerators and currency miners. As users perform workouts on the exercise equipment, their effort is collected as usable electricity and stored in a local grid within the gym. Each watt generated is tracked and associated with the individual who used the machine after a link is created between the machine and an RFID wearable device or smartphone. The credited wattage (stored in the user's "Watt Wallet") can then be spent on tangible rewards within the gym, such as merchandise, vouchers, or even membership discounts. This solution creates a transparent, rewarding feedback loop that incentivizes continued participation while contributing to the gym's sustainability goals.



1.3 Objectives of the Study

This project pursues four primary objectives. First, to design a complete and cohesive system architecture for Power Plant Fitness that illustrates how all components interact within a unified IoT ecosystem. These include components like RFID authentication, exercise equipment, local energy infrastructure, and cloud-hosted services. Second, to produce a minimum of four distinct deliverables: a system architecture diagram, a Python implementation of the reward normalization algorithm, high-fidelity UI mockups for both the machine interface and administrative dashboard, and a written ethics and security analysis. Third, to demonstrate that the proposed system is technically feasible, ethically sound, and achievable within an eight-week academic term using freely available tools and without requiring live software deployment. Fourth, to apply knowledge gained across the MSIT program to a single, cohesive real-world design problem. This includes topics such as IoT systems, human-computer interaction, software engineering principles, and ethics.

Project Goals (SMART Framework)

A common guideline for assessing the realism and feasibility of a project plan is to follow the SMART protocol, which breaks the plan into five measurable criteria that serve as both achievable objectives and performance metrics for progress monitoring (Oguz, 2025). For this project, the following goals will guide the project through the capstone term:

- **Specific:** Design a complete system architecture for Power Plant Fitness that includes all four core components and demonstrates how they interact within a unified IoT ecosystem.
- **Measurable:** Produce a minimum of four distinct deliverables:
 - System architecture diagram
 - Python implementation of the reward normalization algorithm
 - UI mockups for both the machine interface and administrative dashboard
 - Written ethics and security analysis
- **Achievable:** All deliverables are design and documentation artifacts that do not require software development, making them achievable within an eight-week academic term using freely available tools.



- **Relevant:** The project directly addresses a documented problem in the fitness industry and demonstrates mastery of multiple MSIT Program Learning Outcomes, including databases, HCI, software engineering principles, and ethics
- **Time-bound:** All deliverables will be completed by the end of week 6, after which testing, maintenance, documentation, submission, and presentation will happen between weeks 7 and 8.

Expected Outcomes

By the end of this term, the project will produce a comprehensive design and implementation plan for Power Plant Fitness that includes:

- Full system architecture diagram illustrating the data flow between RFID wearables or smartphone NFC, exercise equipment, the centralized backend, and the administrative dashboard
- A relational database schema modeling users, sessions, telemetry data, wattage credits, and product inventory
- High-fidelity mockups of the machine touchscreen interface and administrative portal
- Written analyses covering ethical considerations, security implementations, and project planning

1.4 Scope and Limitations

The deliverables for a project include all products and services created or performed for end users (Oguz, 2025). This project specifically encompasses the design, documentation, and architectural planning of four core system components of Power Plant Fitness:

- An RFID-enabled proximity identification system
- A machine-level touchscreen interface
- A web-based administrative dashboard
- A centralized IT and energy management backend

The deliverables of this project include:

- System architecture diagram



- Python implementation of the reward normalization algorithm
- High-fidelity UI mockups of the machine touchscreen interface and web-based administrative dashboard for staff
- Written documentation covering ethical, security, and planning considerations.

The following are not included in this project's scope:

- Development and deployment of functional software or hardware
- High-fidelity UI mockups of the user-facing mobile app for the gym
- Physical infrastructure specifications, such as network topology or electrical engineering
- Financial planning for the gym construction, equipment acquisition, building leasing, or operations and maintenance

1.5 Report Organization

This report is organized into six main sections. The Introduction to the project includes the background context, problem statement, objectives, and scope. The Literature Review summarizes relevant academic and industry research. It also identifies the gaps in existing work and justifies the proposed system. The System Design describes the full architecture, component specifications, and data flow, and includes a design diagram to illustrate it. The Results of the project include the implementation of the deliverables, interface outputs, and evaluation metrics. The Discussion summarises the findings in the context of existing literature, addresses limitations of the current design, and proposes directions for future work. Finally, the Conclusion summarizes the project's contributions and their implications for the fitness technology industry.



LITERATURE REVIEW

2.1 Overview of Related Work

Gjestvang et al. (2023), in the Journal of Sports and Medicine, studied the characteristics, motivations, and social support of 232 gym members to determine what encourages members to sign up and continue physical training. Despite high interest in joining, dropout rates range from 40% to 65% within the first few months of membership, with motivation and social environment identified as key predictors of sustained engagement (Gjestvang et al., 2023).

Fabrizio et al. (2023) examined new trends and technologies in healthcare and wellness in Healthcare 2023. They determined that while IoT wearables and smart devices have demonstrated measurable benefits for real-time monitoring and personalized feedback, existing platforms primarily focus on self-tracking rather than reward-based systems that incentivize continued participation (Fabrizio et al., 2023).

2.2 Comparative Analysis of Existing Systems

There are several consumer-facing platforms out there that have attempted to address fitness disengagement through digital incentive mechanisms. Numerous smart devices, such as wearables and environmental sensors, use real-time data on users' physical activity to offer personalized training programs and timely feedback, enabling users to monitor their progress, achieve fitness goals, and stay motivated along the way (Fabrizio, 2023). The motivational tactics in these apps focus on achieving goals and maintaining progress, but none of them necessarily reward users while they are actively exercising.

One widely adopted solution to this problem is workout-reward apps like Sweatcoin, Winwalk, CashWalk, Weward, and Walkify. All of these serve the same function of tracking steps and workout metrics to earn a virtual currency that can be redeemed for rewards (Walkify App, n.d.). CashWalk, for example, includes a pedometer that tracks steps, calories burned, and distance, and earns coins that can be redeemed for gift cards (CashWalk Labs, Inc., n.d.). The conceptual



parallel to Power Plant Fitness is clear: all of these apps serve the same purpose: translating physical effort into a redeemable reward currency to sustain long-term engagement.

With all that in mind, the distinction between this solution and workout-reward apps remains significant. The workout-reward apps rely on passive movement tracking or GPS tracking, creating a weak, easily manipulated link between effort and reward. The systems are unable to distinguish between genuine physical exertion and incidental movement, nor do they necessarily account for the activity's intensity. By contrast, Power Plant Fitness uses measurable, physics-based mechanisms to measure physical output accurately and attribute it directly to the individual performing it. The gym also supports a reward system for strength training by integrating the same resistive motors into Leg Press, Leg Extension, Lat Pulldown, Chest Press, and other machines to provide extrinsic motivation for measurable effort.

Additionally, Power Plant Fitness offers not just financial rewards for participation but also ecological incentives. By using generating-type machines, users actively produce real energy that the gym uses to offset external energy demand. This creates an environmental dimension to physical fitness that no other platform has attempted to address.

2.3 Research Gap Identification

As Sheppard (2019) notes, the purpose of a literature review is not only to summarize existing knowledge but to identify research gaps and unanswered questions that justify the need for proposed work. There is a plethora of existing gyms and smart technologies that attempt to help with fitness tracking, reaching personal goals, and providing extrinsic rewards. However, existing gym technology platforms have yet to address the lack of extrinsic reward systems directly tied to their members' physical output (Gjestvang et al., 2023). According to Fabbrizio et al. (2023), there is significant potential for real-time monitoring in fitness environments. However, a clear gap remains: no existing platform links machine-generated energy output to a personalized, redeemable reward economy.

All other consumer-facing workout-reward apps fail to accurately translate real physical effort into redeemable rewards and provide no ecological incentive to participate.

SYSTEM DESIGN

3.1 System Architecture

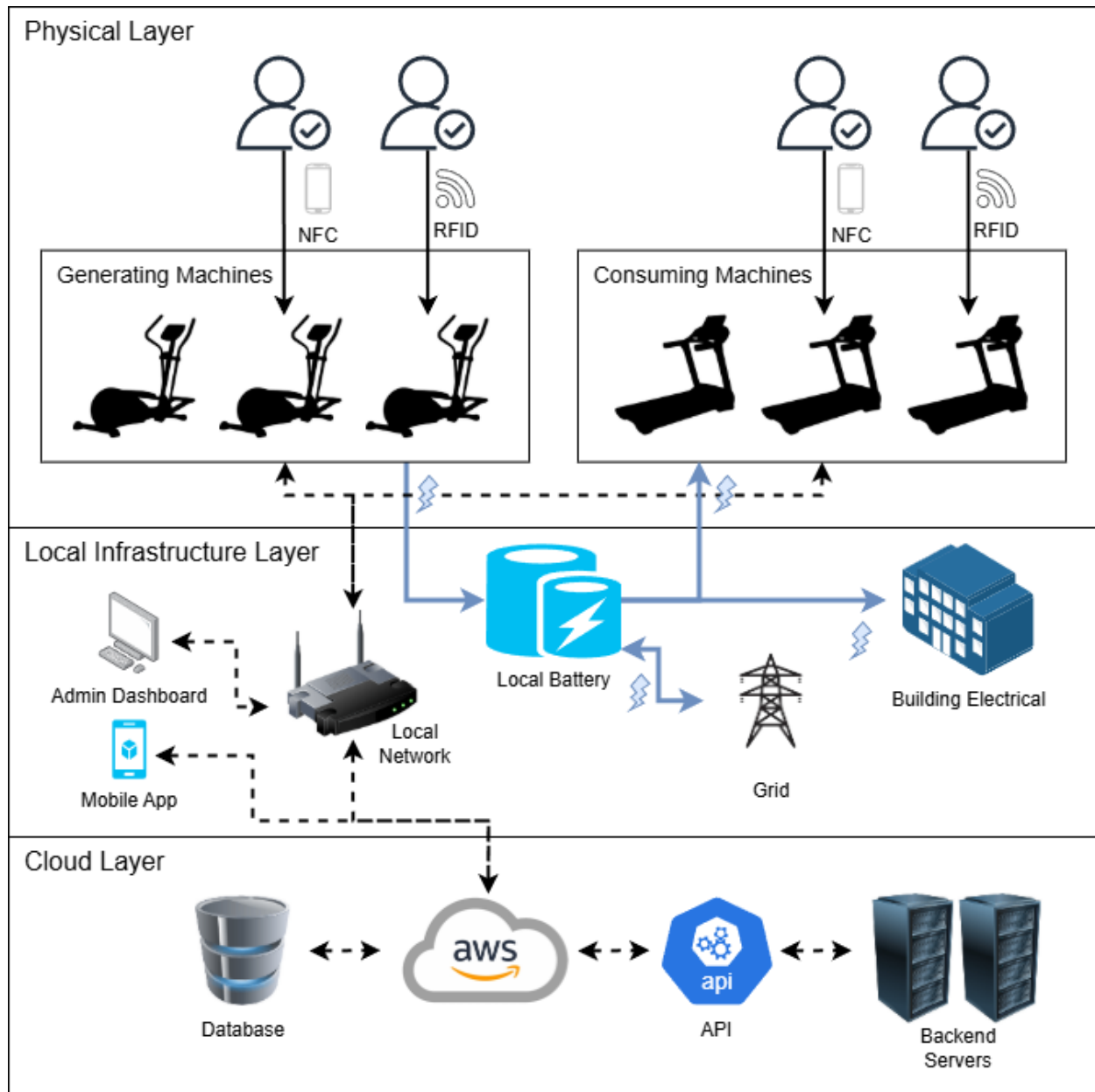


Figure 2. System Architecture Diagram



3.2 Methodologies and Tools Used

This project is primarily focused on the design and documentation of Power Plant Fitness and the IoT infrastructure that defines its core concept. As such, the methodology follows a documentation-driven development approach, in which each project phase produces a distinct design artifact rather than deployable software. Artifacts include things like architecture diagrams, component specifications, interface mockups, and written analyses. The one exception is the reward normalization algorithm, a lightweight Python implementation that serves as a functional proof of concept for the Watt Wallet credit calculation logic. The tools selected for this project reflect this hybrid approach, supporting both design and documentation workflows, as well as a minimal development environment for the algorithm component.

Resource	Type	Justification
Microsoft Word	Documentation	Word Processor for developing the assignments
Draw.io	UI/Diagramming Tool	Versatile drawing tool for creating Gantt charts and system architecture diagrams
Figma	UI Mockup Tool	Industry-standard tool for creating high-fidelity interface mockups
JetBrains PyCharm	Python Development	Integrated Development Environment for developing and testing the reward normalization algorithm for Section 4.
GitHub	Git Repository	A cloud-hosted document and code storage for all project files and versions
AWS Lambda	Serverless Compute	Executes the wattage normalization algorithm in the cloud without provisioning or managing servers, scaling automatically with gym machine activity
AWS API Gateway	REST API Management	Receives HTTP POST requests from gym machines, routes them to the Lambda function, and returns structured JSON responses to update member Watt Wallet balances

Table 1. Resource Justification Table



3.3 Data Flow and System Components

The Power Plant Fitness infrastructure comprises a mix of energy-generating and energy-consuming equipment, touchless authentication mechanisms, a local battery system, administrative and consumer apps, and a centralized cloud-based backend to manage it all. Below is a list of inputs, outputs, and functional and non-functional requirements for each component to prioritize which functionality is important to implement, thereby enabling the project to be more successful and helping developers stay on track (Letaw, 2024).

Energy-Generating Exercise Machines

The most critical components of the gym are the smart exercise machines that use resistive motors to generate electricity. This includes machines like stationary bikes, ellipticals, rowing machines, and cable-based strength machines. Each machine captures mechanical energy based on the user-configured resistance setting during a workout and converts it into usable electricity.

Inputs	Outputs	Methodology	FR	NFR
Physical Effort – torque applied to resistive motor; RFID/NFC session token	Electricity – routed to local grid; session telemetry (like resistance level, RPM, time, etc.)	Energy meters measure real-time wattage output and telemetry statistics and transmit them to the centralized backend.	The machines must accurately measure and transmit wattage during every active session.	Telemetry transmission must occur with a latency under two seconds to ensure accuracy and maintain progress if the connection fails.

Table 2. Energy-generating exercise machine I/O and requirements

Energy-Consuming Exercise Machines

In addition to energy-generating equipment, the gym will be equipped with non-energy-generating machines that will not contribute to redeemable rewards. These include machines like treadmills and stair climbers that draw power from the gym's local batteries instead of contributing to them. These machines will still use RFID authentication to track telemetry statistics and connect to the central server to track energy use, but will not contribute to the user's

wattage balance. A section of the gym may also be sectioned off for free weights, which will not record wattage generation or workout telemetry.

Inputs	Outputs	Methodology	FR	NFR
Electricity – drawn from local battery system; RFID/NFC session token	Energy consumption data; session telemetry (like resistance level, RPM, time, etc.)	Energy meters measure energy consumption and telemetry statistics, and transmit them to the centralized backend	The machines must accurately measure and transmit workout telemetry and energy consumption during every active session	Data monitoring and transmission must not interfere with normal equipment operation

Table 3. Energy-consuming exercise machine I/O and requirements

RFID Wearables and Smartphone NFC

At the time of enrollment at the gym, the user will receive an RFID tag and a silicon wristband with a tab insert. Lanyards would be available at the gym shop. The tag is loaded with an encrypted token associated with the user’s identity, which is then read by a reader embedded in each gym machine. The encryption mitigates spoofing between the tag and reader (Twingate Team, 2024). This will serve as a touchless identification layer that associates each session with a specific member account before any telemetry data is recorded. As an alternative, members can download the Power Plant Fitness mobile app and use their smartphone’s NFC chip to connect to the same readers once logged in.

Inputs	Outputs	Methodology	FR	NFR
Tag RFID signal or smartphone NFC broadcast.	Authenticated session token transmitted to the machine interface and the backend	Mutual authentication ensures that both parties are verified before a session is established.	Authentication must be established before the machine becomes usable	Authentication must be completed within three seconds to avoid lengthy delays

Table 4. RFID wearables and smartphone NFC I/O and requirements



Machine-Level Touchscreen Interface

Every machine is equipped with an interactive touchscreen display that provides real-time session feedback. This includes workout telemetry like elapsed time, selected goal, RPM, resistance level, generated wattage, and cumulative wattage credit. After signing in by tapping their wearable or smartphone to the embedded reader, the screen welcomes them by name and then offers options to configure their workout session and track their progress.

Inputs	Outputs	FR	NFR
RFID/NFC session token; real-time telemetry from onboard energy meter	Visual session data – workout telemetry and wattage generation	Authentication must be established before the machine becomes usable	The interface must be intuitive enough for a first-time user to start a session without assistance (usability)

Table 5. Machine-level touchscreen interface I/O and requirements

Local Battery and Energy Management System

Power Plant Fitness is so named because, in theory, it can generate more energy than it consumes and sell excess power to the grid. A facility-level battery system receives electricity from resistance machines and charges local batteries stored in a secure location within the gym. The gym's local power then discharges the local battery power to enable energy-consuming equipment to operate. A local energy management controller meters both inflow and outflow and logs net energy data to the backend.

Inputs	Outputs	FR	NFR
Electricity – Generated from resistance machines and the outside power grid	Electricity – Required by energy-consuming machines	The system must track net energy generation and consumption in real time, drawing power from the external grid when battery levels are too low	The battery system must maintain stable power delivery to the gym operations regardless of fluctuations in member activity (reliability)

Table 6. Local battery and energy management I/O and requirements

Administrative Dashboard

A browser-based portal accessible to gym staff provides real-time visibility into machine usage, member activity, energy generation metrics, and reward transactions. Role-based access control ensures only authorized individuals can access data relevant to their function.

Inputs	Outputs	FR	NFR
Backend data feeds; staff authentication credentials	Operational reports; manual wattage adjustments	The dashboard must display real-time energy generation totals and member session activity.	The dashboard must be accessible via standard web browsers without requiring additional software installation (usability)

Table 7. Administrative dashboard I/O and requirements

Power Plant Fitness Mobile App

The consumer-facing mobile application allows members to sign in to their accounts remotely, view their session history, monitor their Watt Wallet balance, redeem rewards, participate in personal and group challenges, check leaderboards, and manage their accounts. The app also enables NFC-based machine authentication for members who prefer not to use their RFID tag.

Inputs	Outputs	FR	NFR
Member login credentials; User input for changing configurations, setting goals, and redeeming rewards	Previous session data, Watt Wallet, reward redemption history, leaderboards, challenge updates	The app must allow members to view their data in real-time	The app must be available on both iOS and Android platforms (compatibility)

Table 8. Power Plant Fitness mobile app I/O and requirements

Centralized Cloud Backend

For operational simplicity and flexibility, the backend for the gym's system components will be served and managed on AWS. This will serve as the system's core business logic layer, managing user accounts, processing wattage credit calculations, maintaining the reward ledger, and coordinating data flow among all components. Integrating with AWS allows the gym to



scale up or down with ease, enabling it to pay only for what it needs without requiring infrastructure specialists (GeeksforGeeks, 2025). The backend will also be responsible for normalizing efforts to ensure wattage credits are distributed fairly. Over-rewarding or under-rewarding both affect motivation and purpose, so applying an algorithm to normalize the effort will improve morale and continued participation (Dwivedi, 2024).

Inputs	Outputs	Methodology	FR	NFR
Session telemetry from machines, authentication events, staff, and member app requests	Wattage credit updates; reward transaction records; database feeds; app responses	The reward calculation engine applies a normalized score to ensure equitable credit distribution.	The backend must calculate and apply wattage credits to member accounts within 60 seconds of session completion.	The system must support concurrent sessions across all gym machines without degradation in performance (scalability)

Table 9. Centralized cloud backend I/O and requirements

3.4 Design Diagrams (DFD, UML, ER Diagram)

The system architecture diagram is presented in Section 3.1. Additional diagrams, including a data flow diagram and high-fidelity interface mock-ups for the machine touchscreen and administrative dashboard, are planned deliverables for subsequent project phases. These will be incorporated into the report as they are completed, and will provide further visual clarity on the interaction between system components and the end-user experience.



RESULTS

4.1 Implementation Overview

The main objective of Power Plant Fitness is to encourage member retention and sustain healthy habits by providing a framework that incentivizes participation and rewards effort. To make that process fair and ethical for everyone, the reward system uses a normalization algorithm to calculate the appropriate reward based on the user's level of effort and the consistency they have shown at the gym. The algorithm directly supports the project's central objective of ensuring that the reward system is fair, transparent, and aligned with the gym's primary business goal: keeping users engaged. The rewards generated by the exercise machine feed into a reward loop that provides challenges, goals, and recognition of effort to satisfy and motivate. It should feel like winning a game (Roy, 2025).

The current implementation accepts three inputs per session: the session's current wattage output, the rolling 30-day window used to determine how to reward members based on the number of days in the month they attended, and the member's total lifetime session count. From these, it calculates a consistency bonus of up to 25% applied on top of the member's guaranteed base credit. The idea is to encourage people to keep coming back to the gym rather than focus all their engagement on a few isolated sessions by offering higher rewards for consistency. The bonuses applied after each session help maintain momentum by providing positive reinforcement for consistent effort, helping the user sense that continued motion is more important than a single isolated high point (Alex, 2026).

4.2 Output and Screenshots

The standalone version of this algorithm supports two input modes: a hardcoded demo that runs seven predefined member profiles, and an interactive mode activated via the `--interactive` or `-i` flag that accepts live session input from the command line. In interactive mode, the system prompts the user for three inputs: the wattage generated during the current session, the number of sessions completed in the last 30 days, and the member's total lifetime session count.



Input	
Member name: Samantha Session watts generated: 127 Sessions completed in the last 30 days (not including this one): 7 Total lifetime sessions (not including this one): 7	
Text	
Output	
=====	
Member: Samantha	
=====	
Current Watts:	127.0 Wh
Lifetime Sessions:	7 (Grace Period)
Sessions (Last 30 Days):	7
Consistency Bonus:	1.0
Bonus Multiplier:	1.25
Normalized Credit:	158.75 W
Added To Balance:	158 W
=====	
Demo complete.	
Text	

Figure 3. Input/Output Demo for Samantha

In Figure 3, a gym member, Samantha, has worked on an energy-generating exercise machine, producing 127 Wh of electricity. She has been to this gym 7 times in the last 30 days since she enrolled. Because she has been to the gym fewer than 13 times, she remains in the grace period, which awards her the full 25% bonus regardless of her consistency. This is to ensure that new members are not penalized during the critical early period, when dropout risk is highest.

Input	
Member name: Jacob Session watts generated: 210 Sessions completed in the last 30 days (not including this one): 4 Total lifetime sessions (not including this one): 27	
Text	
Output	
=====	
Member: Jacob	
=====	
Current Watts:	210.0 Wh



Lifetime Sessions:	27
Sessions (Last 30 Days):	4
Consistency Bonus:	0.3333
Bonus Multiplier:	1.0833
Normalized Credit:	227.5 W
Added To Balance:	227 W
=====	
Demo complete.	
Text	

Figure 4. Input/Output Demo for Jacob

In Figure 4, Jacob has generated 210 Wh from his session. The session was his fifth in the last 30 days, but because he has been to the gym more than 12 times, he is no longer in the grace period. Therefore, the consistency bonus drops down to roughly 8% instead of the possible 25% full consistency from 12 sessions per 30 days.

4.3 Evaluation Metrics

Unit Tests

For this project, PyTest was used for unit testing, as it is a widely adopted Python testing framework that excels at supporting both simple unit tests and complex functional testing for applications (GeeksforGeeks, 2026). A total of 15 unit tests were written across five test classes, each targeting a distinct behavioral concern of the `calculate_normalized_credit()` function.

- `testGracePeriod`: validates correct activation and deactivation of the new member grace period
- `testConsistencyBonus`: validates bonus scaling, capping, and edge cases
- `testBaseCredit`: validates that base credit always equals actual watts generated
- `testInputValidation`: validates error handling for invalid inputs
- `testHannahIvanComparison`: validates the core fairness narrative of the system



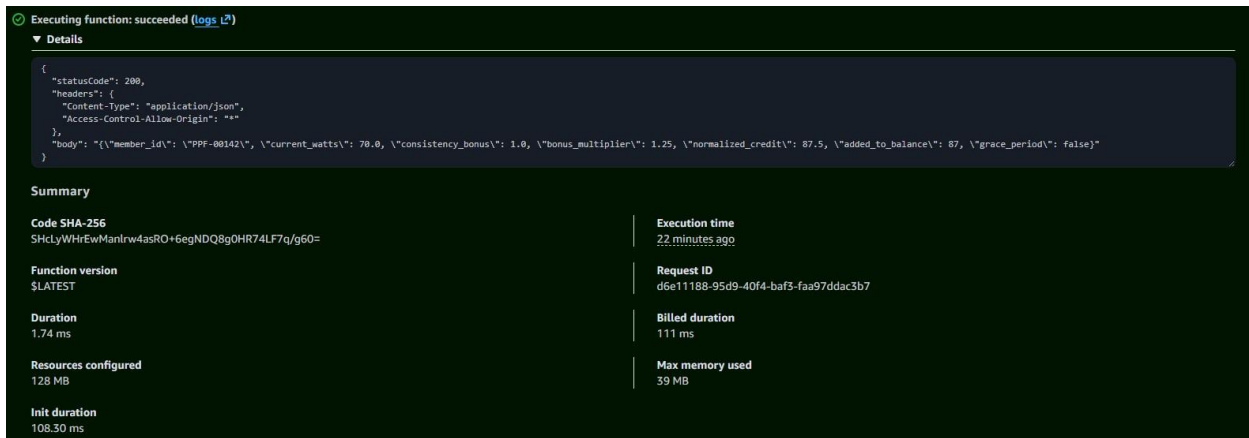
This approach can be characterized as white-box testing, since the tests were written with full knowledge of the internal logic, testing the application's structure and code at the code level (GeeksforGeeks, 2026). Each test targeted a specific branch or calculation within the function. For example, one test verified that the grace period activated at exactly 12 lifetime sessions and deactivated at 13, a result that could only be observed if the tester knew about the internal threshold condition. The results of the test are shown below.

Output
platform win32 -- Python 3.13.0, pytest-9.0.3, pluggy-1.6.0 -- C:\Program Files\Python\Python313\python.exe cachedir: .pytest_cache collecting ... collected 15 items test_watt_calculator.py::TestGracePeriod::test_new_member_receives_full_bonus PASSED [6%] test_watt_calculator.py::TestGracePeriod::test_grace_period_active_at_session_12 PASSED [13%] test_watt_calculator.py::TestGracePeriod::test_grace_period_ends_at_session_13 PASSED [20%] test_watt_calculator.py::TestConsistencyBonus::test_full_bonus_at_target_sessions PASSED [26%] test_watt_calculator.py::TestConsistencyBonus::test_bonus_capped_above_target PASSED [33%] test_watt_calculator.py::TestConsistencyBonus::test_partial_bonus_below_target PASSED [40%] test_watt_calculator.py::TestConsistencyBonus::test_zero_sessions_outside_grace PASSED [46%] test_watt_calculator.py::TestBaseCredit::test_base_credit_guaranteed PASSED [53%] test_watt_calculator.py::TestBaseCredit::test_maximum_bonus_multiplier PASSED [60%] test_watt_calculator.py::TestBaseCredit::test_minimum_bonus_multiplier PASSED [66%] test_watt_calculator.py::TestInputValidation::test_negative_watts_raises_error PASSED [73%] test_watt_calculator.py::TestInputValidation::test_negative_sessions_raises_error PASSED [80%] test_watt_calculator.py::TestInputValidation::test_negative_lifetime_sessions_raises_error PASSED [86%] test_watt_calculator.py::TestInputValidation::test_zero_watts_valid PASSED [93%] test_watt_calculator.py::TestHannahIvanComparison::test_consistent_member_outearn_s_infrequent_per_month PASSED [100%] ===== 15 passed in 0.02s ===== Process finished with exit code 0
Text

Integration Tests

Integration testing was accomplished by attempting to integrate the wattage normalization function into AWS Lambda. AWS Lambda is a compute service that runs without the use of servers, allowing a system or process to run when needed, without requiring an idle server to listen for a request continuously (Amazon Web Services, n.d.). By running my existing `lambda_handler` function on a real Lambda instance, it was verified that the function and the normalization algorithm work together correctly as a unified backend module. 3 tests were conducted on the Lambda instance.

- **Test 1:** Standard session (Hannah)
 - `current_watts`: 70, `sessions_last_30_days`: 20, `total_lifetime_sessions`: 120
 - Results: HTTP 200, `normalized_credit`: 87.5, `grace_period`: false



The screenshot shows the AWS Lambda console interface. At the top, it says 'Executing function: succeeded (logs L2)'. Below this, there's a 'Details' section with a JSON response: `{ "statusCode": 200, "headers": { "Content-Type": "application/json", "Access-Control-Allow-Origin": "*" }, "body": "{\"member_id\": \"PPF-00142\", \"current_watts\": 70.0, \"consistency_bonus\": 1.0, \"bonus_multiplier\": 1.25, \"normalized_credit\": 87.5, \"added_to_balance\": 87, \"grace_period\": false}" }`. Below the details is a 'Summary' section with various metrics:

Summary	
Code SHA-256 5HclyWlrEwManlrw4asRO+6egNDQ8gOHR74LF7q/g60=	Execution time 22 minutes ago
Function version \$LATEST	Request ID d6e11188-95d9-40f4-baf3-faa97ddac3b7
Duration 1.74 ms	Billed duration 111 ms
Resources configured 128 MB	Max memory used 39 MB
Init duration 108.30 ms	

Figure 5. Integration Test 1

- **Test 2:** New member grace period (Samantha)
 - `current_watts`: 127, `sessions_last_30_days`: 7, `total_lifetime_sessions`: 7
 - Results: HTTP 200, `normalized_credit`: 158.75, `grace_period`: true



```
Executing function: succeeded (logs 12)
```

▼ Details

```
{
  "statusCode": 200,
  "headers": {
    "Content-Type": "application/json",
    "Access-Control-Allow-Origin": "*"
  },
  "body": "{\\member_id\\": \\PPF-00001\\", \\current_watts\\": 127.0, \\consistency_bonus\\": 1.0, \\bonus_multiplier\\": 1.25, \\normalized_credit\\": 158.75, \\added_to_balance\\": 158, \\grace_period\\": true}"
}
```

Summary

Code SHA-256	Execution time
Tr0tv4cy32RPxGQIAx5OGS81+BY2B7D4BEe+avfbM=	0 seconds ago
Function version	Request ID
\$LATEST	fcf94957-da1c-4f0d-ba5e-7d98b1f90ac7
Duration	Billed duration
1.59 ms	2 ms
Resources configured	Max memory used
128 MB	40 MB

Figure 6. Integration Test 2

- **Test 3:** Missing field error handling: Submitted payload missing
 - current_watts: 100
 - Results: HTTP 400 with message:
 - Missing required field: 'sessions_last_30_days'

```
Executing function: succeeded (logs 12)
```

▼ Details

```
{
  "statusCode": 400,
  "headers": {
    "Content-Type": "application/json"
  },
  "body": "{\\error\\": \\Missing required field: 'sessions_last_30_days\\'}"
}
```

Summary

Code SHA-256	Execution time
SHcLyWHrEwManlrw4asRO+6egNDQ8g0HR74LF7q/g60=	30 seconds ago
Function version	Request ID
\$LATEST	a1b0f1f9-27be-4f06-8311-cf57cf036e10
Duration	Billed duration
1.46 ms	2 ms
Resources configured	Max memory used
128 MB	40 MB

Figure 7. Integration Test 3

The Lambda execution summary confirmed an actual compute duration of 1.80ms and a maximum memory footprint of 40MB, which was well within the 128MB of memory space allocated. The results also show an initial cold-start time of 108ms, as shown in Figure 2 under “Init Duration”, but this was missing in the following two tests because the Lambda instance was already running at that time. This demonstrates that an execution can be as fast as 1.59ms once the instance is running.



System Tests

System Testing verifies that the entire application is working correctly in a real-world environment (Das, 2025). In this project, system testing means that end-to-end communication from the front end to the back end works as intended. System testing was conducted using Postman, an API client for testing and monitoring Application Programming Interfaces (Caraffa, 2025), by simulating real HTTP POST requests to a live API Gateway endpoint and replicating the exact communication pattern that a Power Plant Fitness machine would use at the end of a workout session. AWS API Gateway is a service for creating, publishing, maintaining, monitoring, and securing REST, HTTP, and Websocket API (Amazon Web Services, n.d.). This was the perfect tool for routing requests to an AWS Lambda, because the integration between the two services is exceptionally straightforward to set up.

The same three integration tests were repeated against the live endpoint. See Figures A1 through A3 in the appendix.

- **Test 1:** Standard session
 - `normalized_credit`: 87.5, HTTP 200, End-to-end response time 392ms (cold start)
- **Test 2:** Grace period member
 - `normalized_credit`: 158.75, `grace_period`: true, HTTP 200, Response time 67ms (warm)
- **Test 3:** Missing field
 - HTTP 400 with a specific missing field identified

The cold start invocation produced an end-to-end response time of approximately 392ms. Subsequent warm invocations reduced this to approximately 67ms, with the normalization algorithm itself executing in under 2ms regardless of cold or warm state. In a production gym

environment, Lambda would remain warm throughout operating hours due to continuous machine activity, making cold starts a non-issue during peak usage.

Algorithm Performance

To assess the raw algorithm's performance, the normalization function was run 1000 times using Python's `time module.perf_counter()` method. The result was 0.0009 milliseconds per calculation, indicating that the engine could theoretically handle over one million sessions per second. In practical deployment during active workout sessions, where the algorithm runs periodically, this performance margin is more than enough to support the entire gym's concurrent memberships without latency issues.

Performance Test	
<pre>... import time ... iterations = 1000 start = time.perf_counter() for _ in range(iterations): calculate_normalized_credit(current_watts=70, sessions_last_30_days=20, total_lifetime_sessions=120) end = time.perf_counter() avg_ms = ((end - start) / iterations) * 1000 print(f"Average execution time: {avg_ms:.4f} ms over {iterations} iterations")</pre>	Python
Output	
Average execution time: 0.0009 ms over 1000 iterations	Text

Figure 8. Algorithm performance test

Qualitative Metric (Interface Usability)

The primary way to interact with exercise machines is through a 24-inch touchscreen interface that controls the machine's operation. The interface was evaluated qualitatively against

fundamental UX heuristics, such as a touch-friendly layout, appropriate scaling, and a minimalist design.

The very first image a user sees when approaching a machine is an instruction to tap their Power Plant Fitness RFID tag on the RFID receiver. This screen's focus is a large illustration of an arm wearing the RFID tag next to the receiver, showing how to follow the instructions. People perceive images faster than words, which helps them understand the context of the message (Jain, n.d.).



Figure 9. Tap-in screen

The home screen presents the member's name, membership tier, watt wallet balance, and real-time energy generation statistics in a single view, matching the mental model a gym member would have of their account without requiring navigation. The buttons are large to make tapping easier, even when you are exhausted and dizzy from an intense workout. Because the screen relies on taps and gestures, the interactions are optimized to be as accessible as possible, which helps ensure users are not frustrated with the UI and increases satisfaction when using the machine, promoting retention (Ndidiama, 2024).

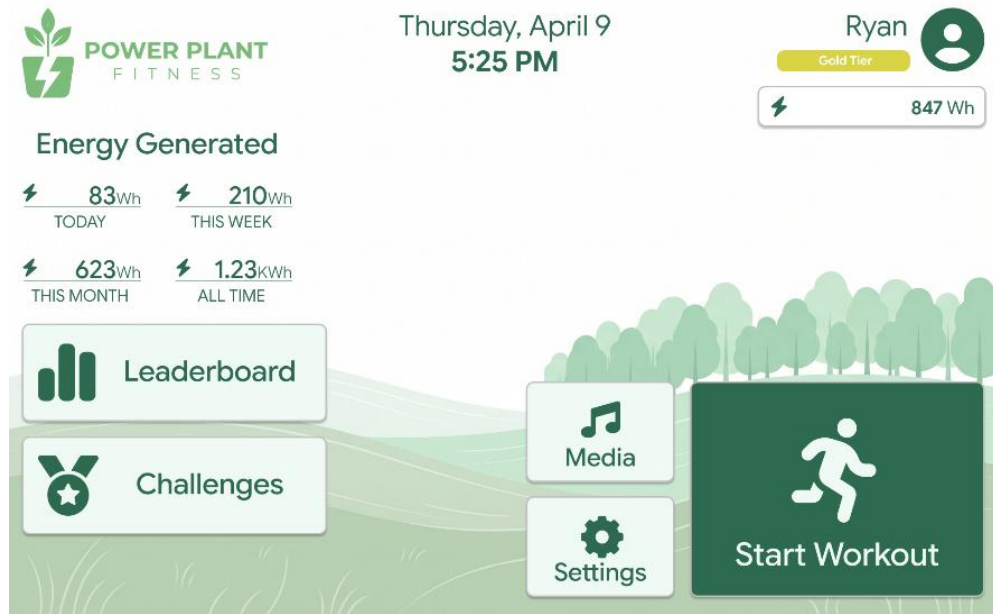


Figure 10. Home screen

Figure 7 demonstrates that the design choices reflect the principle that interface usability is not merely aesthetic; they directly support the system's retention goal by reducing friction at every point of member interaction. Usability ensures that the program is user-friendly and that the end user is happy using the software, and this approach addresses that goal (GeeksforGeeks, 2025).

Results Summary

Figure 8 shows that, while the interface already contributes to a positive user experience, the execution time and test pass rate demonstrate the app's ability to operate quickly and consistently.

Metric	Type	Result	Interpretation
Test pass rate	Quantitative	15/15 (100%)	All functional requirements validated
Algorithm execution time	Quantitative	0.0009 ms/call	Sufficient for real-time gym-scale deployment
Interface usability	Qualitative	Meets 3 UX heuristics	Low friction, brand-consistent, accessible

Figure 11. Evaluation metrics and results

Figure 9 illustrates the monthly Watt Wallet earnings comparison across all seven demo member profiles, with and without the consistency bonus applied. The chart shows that Hannah, attending 20 times per month at 70W per session, outearns Ivan, attending 4 times per month at 350W per session, validating the design intent that consistent attendance is rewarded meaningfully regardless of athletic output.

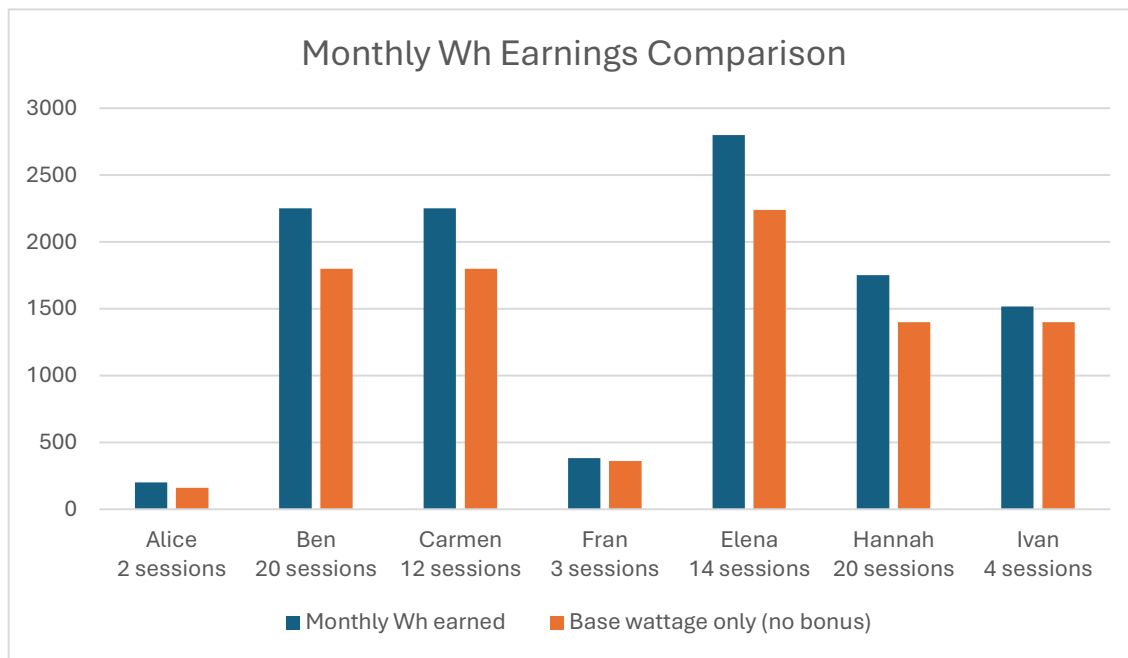


Figure 12. Monthly earnings comparison from demo



DISCUSSION

5.1 Interpretation of Results

The results from the quantitative and qualitative metrics demonstrate that a simple, well-reasoned algorithm and a user-friendly interface can deliver an intuitive, motivational experience that will help ensure member retention. The tests show that the algorithm's logic is functional and consistent: 100% of unit tests across 15 behavioral tests confirm that the normalization engine behaves correctly for normal inputs, edge cases, and invalid inputs. Likewise, the 0.0009ms average execution time confirms that the algorithm introduces no meaningful latency, which is a critical requirement for a system that must update a member's account balance in real time at the end of each session.

The Hannah/Ivan monthly comparison shows an important consideration for health and ethics. The system demonstrated that consistency is rewarded far more than strength and endurance, which is essential for encouraging members to keep coming back and engaging in the gym. Hannah's 20 monthly sessions at 70W yield more total credits than Ivan's 4 sessions at 350W by virtue of a consistency bonus that is rewarded to those who keep coming back. This outcome emerged naturally from the formula when realistic attendance patterns were applied, suggesting the design is genuinely aligned with the retention goal rather than merely approximating it.

The interface usability evaluation reinforces lasting engagement by providing a frictionless tool that users can use to manage their sessions and generate credit in an intuitive, accessible, and visually consistent way that aligns with the brand. A machine touchscreen with these qualities ensures that users can interact with the machines without assistance, which not only boosts morale but also reduces staff demand.

5.2 Comparison with Literature

The project's design decisions are directly supported by the literature reviewed during the initial research phase. Gjestvang et al. (2023) identified dropout rates of 40-65% within the first six months of gym membership, attributing disengagement to insufficient motivation infrastructure



and lack of extrinsic reward mechanisms. The Watt Wallet directly addresses this gap by converting effort into a tangible, redeemable currency, creating an extrinsic feedback loop that existing gym technology platforms do not currently provide.

Rufo et al. (2023) demonstrated that IoT-enabled fitness devices with real-time feedback and gamification features measurably increase user motivation. The Power Plant Fitness architecture implements this IoT feedback loop, extending it with an energy-sustainability dimension that existing platforms have not addressed. The comparison with other gamified exercise apps, like Sweatcoin, Cashwalk, and Walkify, highlights a gap these apps have not addressed: passive monitoring of GPS-based step counts is less verifiable and less physically grounded than Power Plant Fitness's strategy of capturing energy in a more verifiable, physically grounded way.

5.3 Implications and Limitations

Selected Deployment Approach

Several deployment options are viable for Power Plant Fitness, each with distinct tradeoffs in cost, complexity, and scalability. One decision that came to mind was where to host user data and organization software. On-premises infrastructure has many benefits, such as direct control, one-time costs, and internet-free access. However, the added complexity, risks from improper security implementations, and lack of scalability make it an undesirable option (Flower, 2024). Instead, the gym uses AWS CloudFormation to provision an API Gateway and Lambda functions, ensuring simplicity, security, and scalability. This serverless model is particularly well-suited to a gym environment where traffic is already inherently variable. Machine usage would likely peak in the morning and evening, but drop to near zero overnight.

Configuration Management

Software Configuration Management (SCM) is a way to organize and manage project files to increase productivity and reduce errors (Bennett, 2024). In a multi-component system like Power Plant Fitness, where every machine communicates with cloud services and coordinates with other machines, configuration drift across environments is a significant risk. A change to the API endpoint URL, for example, could bring down the entire system. To mitigate this, environment-



specific values, such as API endpoints, database connection strings, and authentication tokens, are stored as environment variables rather than hardcoded in source files. This ensures that the same codebase can be deployed to development, staging, and production environments by simply swapping environment variable values, without modifying any application logic.

Limitations

Several limitations must be acknowledged. From a physical standpoint, not all exercise equipment can generate electricity. Treadmills and stair steppers consume energy rather than produce it, and free weights have no mechanical connection through which energy could be captured. Even resistance-based machines that can generate electricity produce modest amounts. For example, a stationary bike running at a reasonable pace generates approximately 100 Wh, which is roughly enough to power a single lightbulb (Frank, 2016). The intent of the energy generation concept was never to make the gym self-sustainable, but rather to create a meaningful, differentiated incentive for member participation, one in which physical effort carries tangible value beyond personal fitness.

Regarding the project itself, several design assumptions also constitute limitations. The capstone is intended as a design and documentation deliverable: No live software has been deployed, and no real member data has been collected. All outputs are based on simulated profiles that have not been validated against actual gym usage patterns. The system further assumes that manufacturers capable of producing highly integrated, IoT-enabled exercise machines can do so at a viable cost, and that real-world constraints such as regulatory compliance, network reliability, and physical installation costs have been resolved.

5.4 Future Work

Several directions for future development are evident from the current work. The most immediate priority is completing the administrative dashboard and mobile application interface mockups, which were descoped during the capstone term due to time constraints. These components are architecturally defined but not yet visually designed.



Designs should also be completed for the Leaderboards and Challenges screens, which create shared goals for members to pursue and deepen social engagement within the gym community. Weighted consistency targets based on leaderboard rankings could create an aspirational progression system in which Gold Tier members work toward a higher monthly session target than standard members and earn access to superior rewards, thereby incentivizing both retention and tier advancement.

The energy sustainability dimension also merits deeper exploration, as quantifying the actual electricity generation potential of a gym full of resistance machines and modeling the offset against typical gym energy consumption would strengthen the environmental argument and potentially attract sustainability-focused investment. Finally, a live pilot implementation at a real fitness facility would validate the algorithm's fairness assumptions against actual member behavior and provide empirical data on whether consistency bonuses meaningfully improve retention outcomes.



CONCLUSION

The Power Plant Fitness capstone project set out to address one of the fitness industry's most persistent challenges: member retention. Research consistently shows that the majority of gym members disengage within their first six months, not because they lack access to equipment, but because existing motivational infrastructure fails to sustain engagement over time (Gjestvang et al., 2023). This project proposed and designed a system that directly addresses this gap by converting the physical energy members generate during exercise into a redeemable digital currency, and rewarding consistent attendance with a proportional bonus on top of that guaranteed base credit.

The project's primary contribution is the Wattage Credit Normalization Engine, which is a Python-based algorithm that is simultaneously simple, fair, and ungameable. Its simplicity was not a compromise but a deliberate design achievement, achieved through critical evaluation and rejection of more complex approaches that introduced redundancy, compounded penalties, and created gaming vulnerabilities. The algorithm's core principle is that every member always earns at least what they physically generated, with consistency as the only modifier. This goal is defensible, transparent, and communicable to any member in a single sentence.

Several lessons emerged from the development process. First, algorithmic fairness requires asking not just whether a formula produces correct outputs, but whether it produces equitable ones across the full range of members it will serve. The Frank test case, in which a legitimate 120W session yielded only 6 Wh under the original formula, revealed that technical correctness and ethical soundness are distinct requirements that must be explicitly tested. Second, scope management is a genuine skill. Descoping the administrative dashboard and mobile application mockups was the right decision given the timeline, and framing those as future deliverables rather than failures preserved the integrity of the project. Third, version control as documentation, using meaningful commit messages and semantic tags to record design decisions alongside code changes, produced a project history that is itself a readable account of the system's evolution.



The contributions of this project extend beyond its fictional gym setting. The consistency-based reward model, the grace-period mechanism for new members, and the ethical framework for evaluating algorithmic factors are all transferable principles applicable to any IoT-integrated engagement system. As fitness technology continues to evolve and the intersection of sustainability and consumer behavior becomes increasingly relevant, the Power Plant Fitness concept represents a direction worth pursuing: one where physical effort, environmental responsibility, and member well-being are not competing priorities but mutually reinforcing ones.



REFERENCES

- Alex. (2026, April 10). *Understanding reward loops and their impact on player engagement*. Player Counter. <https://playercounter.com/understanding-reward-loops-and-their-impact-on-player-engagement/>
- Amazon Web Services. (n.d.). What is AWS API Gateway? AWS. <https://docs.aws.amazon.com/apigateway/latest/developerguide/welcome.html>
- Amazon Web Services. (n.d.). What is AWS Lambda? AWS. <https://docs.aws.amazon.com/lambda/latest/dg/welcome.html>
- Bennett, L. (2024, August 13). *Software Configuration Management in software engineering*. Guru99. <https://www.guru99.com/software-configuration-management-tutorial.html>
- Caraffa, A. (2025, September 19). Postman: what it is and how it works. <https://openapi.com>. <https://openapi.com/blog/postman-what-it-is-and-how-it-works>
- CashWalk Labs, Inc. (n.d.). *CashWalk – get paid for walking*. CashWalk. Retrieved on April 27, 2026, from <https://cashwalklabs.io/>
- Das, S. (2025, May 29). What is System Testing? (Examples, Use Cases, Types) | BrowserStack. BrowserStack. <https://www.browserstack.com/guide/what-is-system-testing>
- Dwivedi, K. (2024, January 10). What are the most effective ways to balance rewards and incentives? LinkedIn. <https://www.linkedin.com/advice/1/what-most-effective-ways-balance-rewards-incentives-ggufe>
- Fabrizio, A., Fucarino, A., Cantoia, M., De Giorgio, A., Garrido, N. D., Iuliano, E., Reis, V. M., Sausa, M., Vilaça-Alves, J., Zimatore, G., Baldari, C., & Macaluso, F. (2023). *Smart Devices for Health and Wellness Applied to Tele-Exercise: An Overview of New Trends and Technologies Such as IoT and AI*. *Healthcare*, 11(12), 1805. <https://doi.org/10.3390/healthcare11121805>



- Flower, Z. (2024, January 19). *On-premises vs. cloud pros and cons, key differences*. Search Cloud Computing. <https://www.techtarget.com/searchcloudcomputing/tip/Evaluate-on-premises-vs-cloud-computing-pros-and-cons>
- Frank, A. (2016, December 8). *Could you power your home with a bike?* NPR. <https://www.npr.org/sections/13.7/2016/12/08/504790589/could-you-power-your-home-with-a-bike>
- GeeksforGeeks. (2025, August 6). *AWS vs OnPremise Servers: Why Cloud is the Future*. GeeksforGeeks. <https://www.geeksforgeeks.org/blogs/aws-vs-on-premise-servers-why-cloud-is-the-future/>
- GeeksforGeeks. (2025, July 15). *Measuring Software Quality using Quality Metrics*. GeeksforGeeks. <https://www.geeksforgeeks.org/software-engineering/measuring-software-quality-using-quality-metrics/>
- GeeksforGeeks. (2026, March 6). *Pytest Tutorial Unit Testing in Python using Pytest Framework*. GeeksforGeeks. <https://www.geeksforgeeks.org/python/pytest-tutorial-testing-python-application-using-pytest/>
- GeeksforGeeks. (2026, April 28). *White box Testing Software Engineering*. GeeksforGeeks. <https://www.geeksforgeeks.org/software-testing/software-engineering-white-box-testing/>
- Gjestvang, C., Tangen, E. M., Arntzen, M. B., & Haakstad, L. A. (2023). *How do fitness club members differentiate in background characteristics, exercise motivation, and social support? A cross-sectional study*. Journal of sports science & medicine, 22(2), 235. <https://doi.org/10.52082/jssm.2023.235>
- Jain, O. (n.d.). *9 benefits of using illustrations in UI*. Crafttor. Retrieved on May 15, 2026, from <https://crafttor.com/blog/9-benefits-of-using-illustrations-in-ui>



- Letaw, L. (2024). Handbook of software engineering methods [Open textbook]. Oregon State University. <https://open.oregonstate.education/setextbook/> Licensed under CC BY-NC-SA 4.0
- Ndidiamaka, O. (2024, November 30). Touch-Friendly UI design: Best practices to ensure seamless mobile interactions. DEV Community. https://dev.to/okoye_ndidiamaka_5e3b7d30/touch-friendly-ui-design-best-practices-to-ensure-seamless-mobile-interactions-4b1c
- Oguz, A. (2025). *Project Management (2nd ed.)*. MSL Academic Endeavors. <https://pressbooks.ulib.csuohio.edu/projectmanagement2ndedition/> Licensed under CC BY-NC 4.0
- Roy, R. (2025, November 10). *Designing reward loops that keep players hooked (without manipulation)*. <https://medium.com/@rakeshroyakula/designing-reward-loops-that-keep-players-hooked-without-manipulation-58447c858d4a>
- Twingate Team. (2024, August 7). What is RFID spoofing? How it works & examples. Twingate. <https://www.twingate.com/blog/glossary/rfid%20spoofing>
- Walkify App. (n.d.). *Walkify: Walk & get rewards*. Retrieved on April 27, 2026, from <https://www.walkifyapp.com/>



APPENDIX

Figure A1
System Test 1

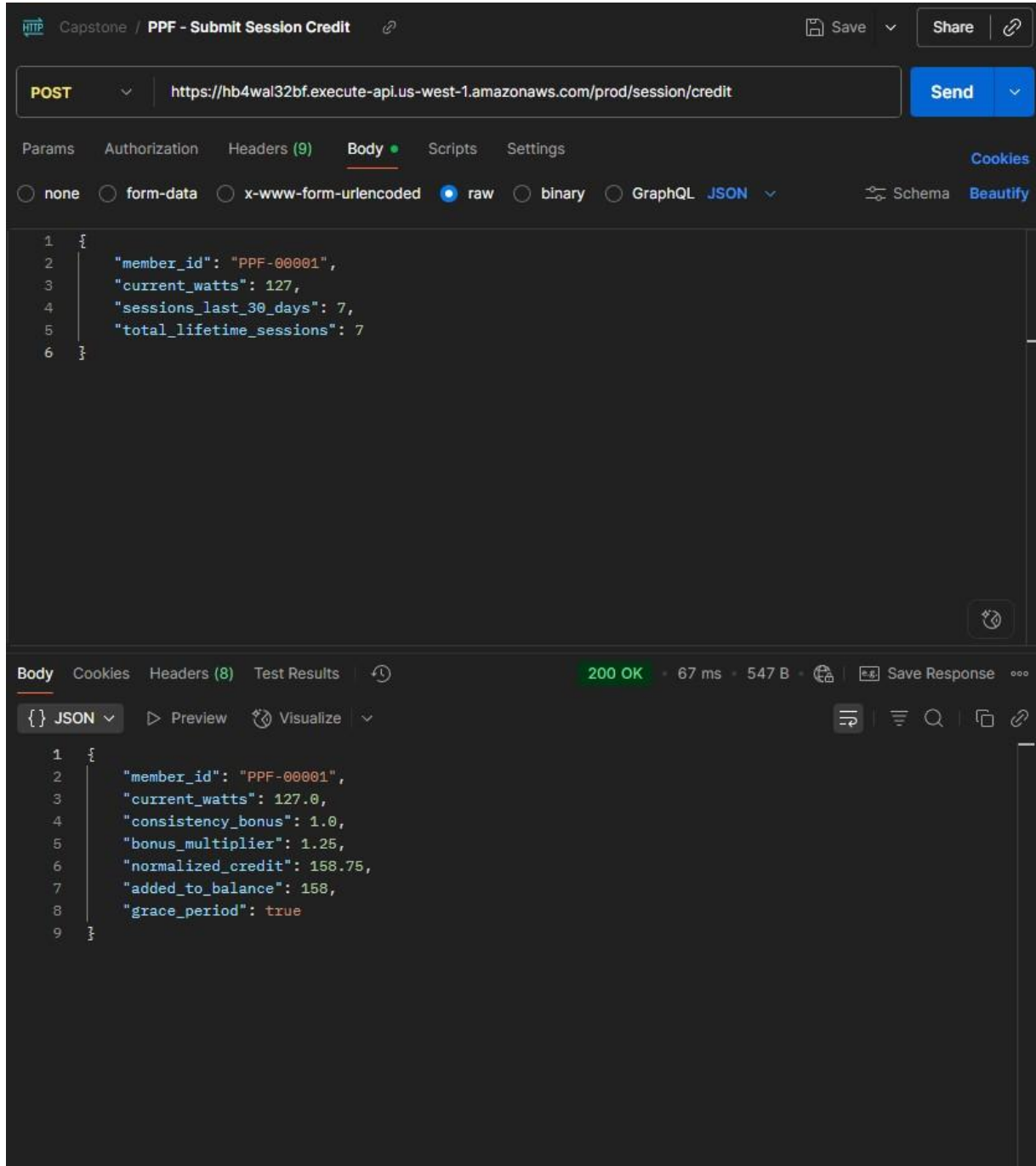
The screenshot displays a REST client interface with the following details:

- URL:** `https://hb4wal32bf.execute-api.us-west-1.amazonaws.com/prod/session/credit`
- Method:** `POST`
- Body (raw):**

```
1 {  
2   "member_id": "PPF-00142",  
3   "current_watts": 70,  
4   "sessions_last_30_days": 20,  
5   "total_lifetime_sessions": 120  
6 }
```
- Response Status:** `200 OK` (392 ms, 544 B)
- Response Body (JSON):**

```
1 {  
2   "member_id": "PPF-00142",  
3   "current_watts": 70.0,  
4   "consistency_bonus": 1.0,  
5   "bonus_multiplier": 1.25,  
6   "normalized_credit": 87.5,  
7   "added_to_balance": 87,  
8   "grace_period": false  
9 }
```

Figure A2
System Test 2



The screenshot displays a REST client interface for a POST request. The URL is `https://hb4wal32bf.execute-api.us-west-1.amazonaws.com/prod/session/credit`. The request body is a JSON object with the following fields: `member_id` (PPF-00001), `current_watts` (127), `sessions_last_30_days` (7), and `total_lifetime_sessions` (7). The response is a 200 OK status with a 67 ms latency and 547 B of data. The response body is a JSON object with the following fields: `member_id` (PPF-00001), `current_watts` (127.0), `consistency_bonus` (1.0), `bonus_multiplier` (1.25), `normalized_credit` (158.75), `added_to_balance` (158), and `grace_period` (true).

Capstone / PPF - Submit Session Credit

POST `https://hb4wal32bf.execute-api.us-west-1.amazonaws.com/prod/session/credit` Send

Params Authorization Headers (9) Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```
1 {
2   "member_id": "PPF-00001",
3   "current_watts": 127,
4   "sessions_last_30_days": 7,
5   "total_lifetime_sessions": 7
6 }
```

Body Cookies Headers (8) Test Results 200 OK • 67 ms • 547 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "member_id": "PPF-00001",
3   "current_watts": 127.0,
4   "consistency_bonus": 1.0,
5   "bonus_multiplier": 1.25,
6   "normalized_credit": 158.75,
7   "added_to_balance": 158,
8   "grace_period": true
9 }
```



Figure A3
System Test 3

The screenshot shows a REST client interface with the following details:

- URL:** `https://hb4wal32bf.execute-api.us-west-1.amazonaws.com/prod/session/credit`
- Method:** `POST`
- Body (JSON):**

```
1 {
2   "member_id": "PPF-00999",
3   "current_watts": 100
4 }
```
- Response:** `400 Bad Request` (29 ms, 405 B)
- Response Body (JSON):**

```
1 {
2   "error": "Missing required field: 'sessions_last_30_days'"
3 }
```